AUV SIMULATION SCENARIO SPECIFICATIONS

Comprehensive Technical Audit of Actuators, Sensors, Hydrodynamics & Code Placeholders

SCENARIO FILE	SIMULATOR	PHYSICS FRAME	VEHICLE MASS / DEPTH
auv_scenario.xml (v10)	Stonefish ROS 2	NED (+Z Down, SI Units)	156.0 kg / 10.0 m Seabed

8 Thrusters
4 HORIZ VECTORED + 4 VERTICAL

4 Active
SENSORS (IMU, DVL, DEPTH, SONAR)

1 Missing
ZED CAMERA (V10 PLACEHOLDER)

1 Pending
CG OFFSET (SOLIDWORKS CAD)

1 Vehicle Physical & Hydrodynamic Architecture

The simulated vehicle is modeled as a compound submerged rigid body with buoyancy dynamics in North-East-Down (NED) coordinates (**+X Surge / Forward**, **+Y Sway / Starboard**, **+Z Heave / Downward**).

Hull & Rigid Body Variables Dictionary

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & SIGNIFICANCE
`base_link name`	`"Vehicle"`	–	Root dynamic link representing the AUV rigid core to which all actuators and sensors attach.
`base_link physics`	`"submerged"`	–	Enables Stonefish fluid-drag and hydrostatic buoyancy computation for links immersed in water.
`external_part buoyant`	`"true"`	–	Instructs simulator to tetrahedralize collision mesh to calculate submerged volume & Center of Buoyancy (CB).
`physical mesh`	`"auv_hull_phys.obj"`	–	Low-poly watertight 3D mesh used by physics engine for collision checks and volumetric buoyancy integration.
`visual mesh`	`"auv_hull.obj"`	–	High-fidelity 3D mesh rendered visually with PBR materials in the OpenGL simulator window.
`mass value`	`"156"`	kg	Total dry vehicle mass ($156.0\ \text{kg}$). Multiplied by gravity g to determine gravitational downward force ($W \approx 1530.4\ \text{N}$).
`inertia xyz`	`"3.928 3.861 1.164"`	kg·m²	Diagonal moments of inertia: $I_{xx} = 3.928$ (Roll), $I_{yy} = 3.861$ (Pitch), $I_{zz} = 1.164$ (Yaw). Low I_{zz} enables rapid yaw rotation.
`world_transform xyz`	`"0.0 0.0 2.0"`	m	Initial spawn position in NED world frame. $z = 2.0\ \text{m}$ spawns the vehicle $2.0\ \text{m}$ below the surface ($8.0\ \text{m}$ above seabed).

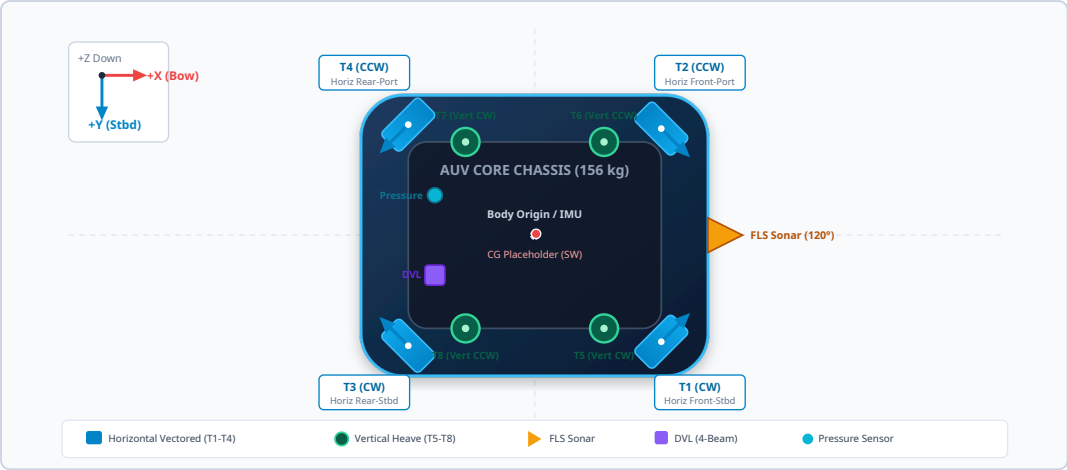
Environment & Ocean Parameters Dictionary

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & SIGNIFICANCE
`ned latitude / longitude`	`"30.0" / "31.0"`	deg	Geographic origin anchor point (Cairo, Egypt). Used by simulated GPS/INS to convert local NED meters to WGS84.
`ocean waves height`	`"0.0"`	m	Surface wave amplitude. Kept at 0.0 to prevent OpenGL compute shader crashes on non-NVIDIA GPUs.
`current velocity`	`"0.0 0.0 0.0"`	m/s	Uniform ocean water current vector $[u_x, u_y, u_z]$. Currently quiescent; can be adjusted for disturbance rejection tests.
`static Bottom transform`	`"0.0 0.0 10.0"`	m	Horizontal rock seabed plane located at $z = 10.0\ \text{m}$ depth. Defines bottom altitude boundary for DVL tracking.

CRITICAL CODE PLACEHOLDER: Center of Gravity (CG) Offset Commented Out
In line 64 of `auv_scenario.xml`: `<!-- <cg xyz="0.00281 -0.00154 -0.3092" rpy="0 0 0"/> -->`.
What it means: Omitting `<cg>` sets the Center of Mass (CM) at geometric origin $(0,0,0)$. The CAD value has $z = -0.3092\ \text{m}$ (above origin in NED). If the Center of Buoyancy (CB) is below this point, metacentric height $\overline{GM} < 0$, causing capsizing. Verify ballast before uncommenting.

2 8-Thruster & Sensor Vector Geometry (Top View)

The AUV employs an 8-thruster arrangement combining four horizontal diagonal vectored thrusters (T1–T4) with four vertical heave thrusters (T5–T8) in the top plate corner holes.



3 Thruster / Actuator Variables Master Dictionary

The table below explains every single parameter defined inside the `<actuator type="thruster">` XML blocks and its physical effect on simulation dynamics:

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & PHYSICAL EFFECT
`actuator name`	`"ThrusterT1" .. T8`	–	Unique identifier for the actuator within the robot kinematic tree. Used by ROS 2 setpoint dispatcher.
`specs max_setpoint`	`"100.0"`	rad/s	Maximum rotational angular velocity of the propeller (ω_max = 100 rad/s ≈ 955 RPM).
`specs normalized_setpoint`	`"true"`	boolean	When true, ROS command input values are expected in normalized range [-1.0, 1.0]. Input 1.0 yields ω_max.
`specs inverted_setpoint`	`"false"`	boolean	If true, inverts algebraic sign of commanded setpoint. Kept false since chirality handles direction.
`propeller diameter`	`"0.1"`	m	Propeller blade tip diameter (D = 10.0 cm). Used for disc area calculations and mesh scaling.
`propeller right`	`"true" / "false"`	boolean	Chirality / handedness: `true` = Right-handed (CW), `false` = Left-handed (CCW). Crucial for torque balancing.
`rotor_dynamics time_constant`	`"0.1"`	s	First-order lag τ = 0.1 s. Propeller reaches 63.2% of commanded RPM in 100 ms, modeling motor inertia.
`thrust_coeff forward`	`"0.88"`	N/(rad/s)²	Forward thrust factor Ct in T = Ct · ω|ω|. Generates 0.88 N at ω = 1 rad/s (8800 N at max ω = 100 rad/s).
`thrust_coeff reverse`	`"0.88"`	N/(rad/s)²	Reverse thrust factor. Set symmetric to forward in current XML (placeholder for real propeller asymmetry).
`torque_coeff value`	`"0.05"`	N·m/(rad/s)²	Hydrodynamic drag counter-torque Q = Cq · ω|ω| transmitted back into AUV body around shaft axis.
`origin xyz / rpy`	`(varies per thruster)`	m / rad	Mounting 3D offset and rotation. Thrust vector is directed along the local thruster $+X$ axis.

3.1 Individual Thruster Parameters Breakdown (T1 to T8)

Detailed positional and angular parameters for all eight thrusters. Roll, pitch, and yaw angles are expressed in both radians and degrees:

ID	PHYSICAL ROLE & LOCATION	MOUNTING ORIGIN [M]	ORIENTATION (RPY)	CHIRALITY	THRUST VECTOR DIRECTION	PRIMARY DOF CONTROLLED

T1	Horizontal Front-Starboard	`X: +0.2537`
		`Y: +0.2213`
		`Z: +0.0227`
			`R: 0.0 rad (0°)`
			`P: 0.0 rad (0°)`
			`Y: -0.7854 rad (-45°)`
				RIGHT (CW)	$[+0.7071, -0.7071, 0.0]$
					Forward & Port push
						Surge ($+F_x$), Sway ($-F_y$), Yaw ($-M_z$)

T2	Horizontal Front-Port	`X: +0.2537`
		`Y: -0.2213`
		`Z: +0.0227`
			`R: 0.0 rad (0°)`
			`P: 0.0 rad (0°)`
			`Y: +0.7854 rad (+45°)`
				LEFT (CCW)	$[+0.7071, +0.7071, 0.0]$
					Forward & Stbd push
						Surge ($+F_x$), Sway ($+F_y$), Yaw ($+M_z$)

T3	Horizontal Rear-Starboard	`X: -0.2463`
		`Y: +0.2287`
		`Z: +0.0227`
			`R: 0.0 rad (0°)`
			`P: 0.0 rad (0°)`
			`Y: +3.9270 rad (+225°)`
				RIGHT (CW)	$[-0.7071, -0.7071, 0.0]$
					Aft & Port push
						Surge ($-F_x$), Sway ($-F_y$), Yaw ($-M_z$)

T4	Horizontal Rear-Port	`X: -0.2463`
		`Y: -0.2287`
		`Z: +0.0227`
			`R: 0.0 rad (0°)`
			`P: 0.0 rad (0°)`
			`Y: -3.9270 rad (-225°)`
				LEFT (CCW)	$[-0.7071, +0.7071, 0.0]$
					Aft & Stbd push
						Surge ($-F_x$), Sway ($+F_y$), Yaw ($+M_z$)

T5	Vertical Front-Starboard	`X: +0.1430`
		`Y: +0.1945`
		`Z: -0.0565`
			`R: 0.0 rad (0°)`
			`P: -1.5708 rad (-90°)`
			`Y: 0.0 rad (0°)`
				RIGHT (CW)	$[0.0, 0.0, +1.0]$
					Pushes Downward (+Z)
						Heave ($+F_z$), Pitch ($-M_y$), Roll ($+M_x$)

T6	Vertical Front-Port	`X: +0.1430`
		`Y: -0.1945`
		`Z: -0.0565`
			`R: 0.0 rad (0°)`
			`P: +1.5708 rad (+90°)`
			`Y: 0.0 rad (0°)`
				LEFT (CCW)	$[0.0, 0.0, -1.0]$
					Pushes Upward (-Z)
						Heave ($-F_z$), Pitch ($+M_y$), Roll ($+M_x$)

T7	Vertical Rear-Port	`X: -0.1430`
		`Y: -0.1945`
		`Z: -0.0565`
			`R: 0.0 rad (0°)`
			`P: -1.5708 rad (-90°)`
			`Y: 0.0 rad (0°)`
				RIGHT (CW)	$[0.0, 0.0, +1.0]$
					Pushes Downward (+Z)
						Heave ($+F_z$), Pitch ($+M_y$), Roll ($-M_x$)

T8	Vertical Rear-Starboard	`X: -0.1430`
		`Y: +0.1945`
		`Z: -0.0565`
			`R: 0.0 rad (0°)`
			`P: +1.5708 rad (+90°)`
			`Y: 0.0 rad (0°)`
				LEFT (CCW)	$[0.0, 0.0, -1.0]$
					Pushes Upward (-Z)
						Heave ($-F_z$), Pitch ($-M_y$), Roll ($-M_x$)

Motor Lag & Thrust Mathematical Equations

1st-Order Electro-Mechanical Lag:
$\tau \cdot d\omega/dt + \omega = \omega_{cmd}$ (with time constant $\tau = 0.10$ s)
Hydrodynamic Thrust & Reaction Torque Law:
Thrust: $T = Ct \cdot \omega|\omega| = 0.88 \cdot \omega|\omega|$ [N]
Reaction Torque: $Q = Cq \cdot \omega|\omega| = 0.05 \cdot \omega|\omega|$ [N·m]

ROS 2 Command Subscription Interface

Subscriber Topic: `/auv/thruster_setpoints`
Message Type: `std_msgs/msg/Float64MultiArray`
Array Format: Float vector with 8 elements: [u1, u2, u3, u4, u5, u6, u7, u8].
Input Range: [-1.0, 1.0]. Value 0.0 stops motor; 1.0 spins at $\omega_{max} = 100$ rad/s.

Thrust Allocation Matrix (TAM) Mapping Matrix B

Maps 6-DOF force/torque vector $\tau = [Fx, Fy, Fz, Mx, My, Mz]^T$ to 8 thruster setpoints ($\tau = B \cdot u$):

```
# Rows:    [ Surge (Fx), Sway (Fy), Heave (Fz), Roll (Mx), Pitch (My), Yaw (Mz) ]
# Columns: [ T1,         T2,        T3,         T4,        T5,         T6,        T7,         T8 ]
[  0.7071,  0.7071, -0.7071, -0.7071,     0.0,       0.0,       0.0,       0.0   ]  # Fx (Surge)
[ -0.7071,  0.7071, -0.7071,  0.7071,     0.0,       0.0,       0.0,       0.0   ]  # Fy (Sway)
[     0.0,     0.0,     0.0,     0.0,  1.0000,   -1.0000,    1.0000,   -1.0000   ]  # Fz (Heave)
[     0.0,     0.0,     0.0,     0.0,  0.1945,    0.1945,   -0.1945,   -0.1945   ]  # Roll (Mx)
[     0.0,     0.0,     0.0,     0.0, -0.1430,    0.1430,    0.1430,   -0.1430   ]  # Pitch (My)
[ -0.3359,  0.3359, -0.3359,  0.3359,     0.0,       0.0,       0.0,       0.0   ]  # Yaw (Mz)
```

4 IMU & DVL Sensor Variables Deep-Dive

Exhaustive parameter breakdown for the Inertial Measurement Unit (IMU) and Doppler Velocity Log (DVL):

Sensor 1: Inertial Measurement Unit (<sensor name="IMU" type="imu">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"50.0"`	Hz	Output frequency. Publishes 50 messages/sec (every 20 ms), providing high-rate attitude dynamics for EKF/Madgwick filters without aliasing.
`range angular_velocity`	`"10.0 10.0 10.0"`	rad/s	Full-scale gyro measurement ceiling on X, Y, Z axes (±10.0 rad/s ≈ ±573°/s). Higher angular velocities will saturate digitally.
`range linear_acceleration`	`"20.0"`	m/s²	Full-scale accelerometer measurement ceiling (±20.0 m/s² ≈ ±2.04 g). Accurately captures gravity vector plus vehicle surge/sway/heave.
`noise angle`	`"0.001 0.001 0.001"`	rad	White Gaussian noise standard deviation on orientation quaternion (1.0 mrad ≈ 0.057°). Simulates internal AHRS fusion uncertainty.
`noise angular_velocity`	`"0.0001"`	rad/s	Gyroscope noise standard deviation (0.0001 rad/s ≈ 0.0057°/s). Tactical-grade MEMS gyroscope noise density.
`noise yaw_drift`	`"0.0001"`	rad/s	Continuous uncompensated heading bias drift rate. Simulates gyro random walk in the absence of absolute magnetometer aiding.
`noise linear_acceleration`	`"0.001"`	m/s²	Accelerometer measurement white noise standard deviation (0.001 m/s² ≈ 0.1 mg). Represents high-precision accelerometer output.
`history samples`	`"1"`	samples	Internal filter queue depth. 1 sample outputs instantaneous raw measurements with zero moving-average group delay.
`origin xyz / rpy`	`"0.0 0.0 0.0" / "0 0 0"`	m / rad	Mounted exactly at body geometric center. Eliminates parasitic centrifugal and tangential acceleration offsets (ω × (ω × r)).
`ros_publisher topic`	`"/auv/imu"`	–	Target ROS 2 topic emitting standard message type `sensor_msgs/msg/Imu`.

Sensor 2: Doppler Velocity Log (<sensor name="DVL" type="dvl">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"5.0"`	Hz	Acoustic ping update frequency. Emits 5 velocity packets per second (every 200 ms), typical for subsea acoustic travel time over depth.
`specs beam_angle`	`"30.0"`	deg	Elevation tilt angle of the 4 Janus transducer beams relative to vertical axis (30.0°). Used by Janus matrix to resolve 3D body velocity.
`range velocity`	`"3.0 3.0 1.0"`	m/s	Maximum measurable speed-over-ground: ±3.0 m/s in Surge (X), ±3.0 m/s in Sway (Y), and ±1.0 m/s in Heave (Z).
`range altitude_min / max`	`"0.5" / "50.0"`	m	Operational bottom-lock window: minimum blanking zone of 0.5 m (near-field transducer ringing) up to 50.0 m maximum bottom track.
`noise velocity`	`"0.0015"`	m/s	Velocity measurement noise standard deviation (1.5 mm/s RMS). Provides ultra-low drift velocity aiding for dead reckoning.
`noise altitude`	`"0.001"`	m	Acoustic altimeter range noise standard deviation (1.0 mm RMS). Millimeter-precision distance to seabed.
`history samples`	`"1"`	samples	No internal sliding window averaging applied; publishes single-ping instantaneous Doppler shifts.
`origin xyz / rpy`	`"-0.19 0.19 0.45" / "0 0 0"`	m / rad	Mounted on aft-starboard underside (x = -0.19 m, y = +0.19 m, z = +0.45 m), pointing directly downwards into water column.
`ros_publisher topic`	`"/auv/dvl"`	–	Publishes `stonefish_ros2/msg/DVL` (velocity vector, beam ranges, validity flags). Optional altitude published to `sensor_msgs/Range`.

4.1 Pressure Sensor, FLS Sonar & Camera Variables Deep-Dive

Exhaustive parameter breakdown for Subsea Pressure Sensor, Forward-Looking Imaging Sonar (FLS), and ZED Camera:

Sensor 3: Hydrostatic Depth / Pressure (<sensor name="Pressure" type="pressure">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"10.0"`	Hz	Output frequency. Pressure readings emitted 10 times per second (every 100 ms) for closed-loop depth holding.
`range pressure`	`"1000000.0"`	Pa	Maximum pressure limit (10^6 Pa = 10 bar). Derived depth: $z \approx (P - P_0) / (\rho \cdot g) \approx 91.7$ m subsea operational depth rating.
`noise pressure`	`"10.0"`	Pa	Hydrostatic pressure noise standard deviation (10.0 Pa). Equivalent to millimeter depth resolution: $\Delta z \approx 10 / (1000 \times 9.81) \approx 1.02$ mm H_2O.
`history samples`	`"1"`	samples	Single sample history without lag; delivers instantaneous pressure samples.
`origin xyz / rpy`	`"-0.19 0.19 0.20" / "0 0 0"`	m / rad	Port tap position located 0.20 m below body geometric center (x = -0.19 m, y = +0.19 m).
`ros_publisher topic`	`"/auv/pressure"`	–	Target ROS 2 topic emitting standard message type `sensor_msgs/msg/FluidPressure`.

Sensor 4: Forward-Looking Imaging Sonar (<sensor name="FLS" type="fls">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"10.0"`	Hz	Acoustic frame rate. Generates 10 complete 2D acoustic echogram images per second.
`specs beams`	`"512"`	beams	Horizontal acoustic receiver beams across aperture. High angular resolution: $\Delta\theta = 120° / 512 \approx 0.234°$ per beam.
`specs bins`	`"500"`	bins	Radial range sample cells per beam. Fine range resolution: $\Delta r = (10.0 - 0.5) / 500 = 1.9$ cm per cell.
`specs horizontal_fov`	`"120.0"`	deg	Horizontal azimuthal aperture opening angle (120.0°). Provides wide-angle forward situational awareness.
`specs vertical_fov`	`"120.0"`	deg	Vertical beam width cone (120.0°). Illuminates from water surface down to seabed.
`specs output_format`	`"uint8"`	–	8-bit unsigned integer per pixel (0 - 255). 0 = no acoustic backscatter (clear water), 255 = maximum specular reflection.
`settings range_min / max`	`"0.5" / "10.0"`	m	Active acoustic imaging range: 0.5 m near-field blind zone up to 10.0 m maximum obstacle detection distance.
`settings gain`	`"1.1"`	–	Acoustic receiver amplification factor (1.1x). Boosts weak echoes from distant or low-reflectivity targets.
`noise multiplicative / add`	`"0.01" / "0.02"`	–	Acoustic clutter: 1% multiplicative speckle noise plus 2% additive ambient water noise.
`display colormap`	`"perula"`	–	Color lookup table for Stonefish visualizer GUI rendering (blue-green-yellow high-contrast palette).
`origin xyz / rpy`	`"0.35 0.0 -0.10" / "0 1.5708 0"`	m / rad	Mounted at bow (x = +0.35 m, z = -0.10 m) pitched by 90° (1.5708 rad) to aim acoustic cone directly ahead.
`ros_publisher topic`	`"/auv/display"`	–	Publishes raw 2D polar echogram matrix as standard ROS 2 `sensor_msgs/msg/Image`.

Sensor 5: ZED 2i Stereo Camera (Planned / Header Placeholder)

PARAMETER / XML ATTRIBUTE	RECOMMENDED VALUE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"30.0"`	Hz	Real-time video frame rate (30 FPS) for visual odometry (VIO) and YOLO deep learning object detection.
`specs resolution`	`1280 x 720`	pixels	HD 720p color video stream providing high feature density without exceeding GPU rendering bandwidth.
`specs horizontal_fov`	`"110.0"`	deg	Wide field-of-view matching the physical Stereolabs ZED 2i optical lens underwater profile.
`origin xyz / rpy`	`"0.32 0.0 0.05" / "0 0 0"`	m / rad	Bow forward mount (x = +0.32 m, deck level z = +0.05 m) pointing directly along vehicle +X heading.
`ros_publisher topic`	`"/auv/zed/image_raw"`	–	Emits `sensor_msgs/msg/Image` and accompanying camera calibration matrix `sensor_msgs/msg/CameraInfo`.

5 Code Placeholders & Architectural Recommendations

1. MISSING HARDWARE SENSOR: ZED Stereo Camera Tag
Audit Finding: Line 5 header declares: `v10: + DVL, pressure, 1D sonar, ZED.` However, no camera sensor tag exists in lines 216–255! Vision pipelines (YOLO gate/buoy detection, VIO, visual SLAM) cannot launch without camera topics.

2. UNENABLED TELEMETRY: Thruster State Feedback Publisher
Audit Finding: Line 256 defines `<ros_subscriber thrusters="/auv/thruster_setpoints"/>`, but lacks thruster state telemetry feedback.
Enhancement: Adding `<ros_publisher thrusters="/auv/thruster_state"/>` exposes real-time propeller RPM, actual thrust force, and torque feedback.

3. GPU CRASH SAFEGUARD: Ocean Wave Height Locked at 0.0
Technical Context: Line 14 sets `<waves height="0.0"/>` with the comment `Keep 0.0: any wave height crashes the shaders on your GPU.` Stonefish wave surface rendering uses OpenGL compute shaders. Keep 0.0 when running on integrated GPUs or software Mesa drivers.

4. OCEAN CURRENT DISTURBANCE & SENSOR SERVICE CONTROLS
• **Current Velocity:** `<velocity xyz="0.0 0.0 0.0"/>` is currently quiescent. Inject non-zero values (e.g. 0.25 m/s) to benchmark disturbance rejection.
• **Dynamic Power Services:** Add `<ros_service set_enabled="/auv/sensor_name/set_enabled"/>` to toggle heavy sensors like the 512-beam FLS sonar via `std_srvs/srv/SetBool`.

6 Ready-to-Use XML Drop-in Code Snippets

A. ZED 2i Stereo Camera (Insert before <ros_subscriber>)

```
<!-- ================= ZED 2i STEREO CAMERA ================= -->
<sensor name="ZED_Camera" rate="30.0" type="camera">
    <specs resolution_x="1280" resolution_y="720" horizontal_fov="110.0"/>
    <origin xyz="0.32 0.0 0.05" rpy="0.0 0.0 0.0"/>
    <link name="Vehicle"/>
    <ros_publisher topic="/auv/zed/image_raw"/>
    <ros_service set_enabled="/auv/zed/set_enabled"/>
</sensor>
```

B. Thruster State Feedback & Verified Center of Gravity

```
<!-- Enable real-time telemetry of thruster RPM, thrust force, and torque -->
<ros_publisher thrusters="/auv/thruster_state"/>

<!-- Verified SolidWorks CG (Uncomment after checking ballast stability GM > 0) -->
<cg xyz="0.00281 -0.00154 -0.3092" rpy="0.0 0.0 0.0"/>
```

7 Master ROS 2 Topics & Interfaces Summary

TOPIC NAME	DIRECTION	MESSAGE TYPE	RATE	DESCRIPTION
`/auv/thruster_setpoints`	SUB	`std_msgs/msg/Float64MultiArray`	Async	8-element vector of normalized thruster setpoints [-1.0, 1.0]
`/auv/thruster_state`	PUB	`stonefish_ros2/msg/ThrusterState`	Sync	Real-time feedback: setpoints, angular velocities (rad/s), thrusts (N)
`/auv/imu`	PUB	`sensor_msgs/msg/Imu`	50 Hz	Linear accelerations, angular rates, and fused orientation quaternion
`/auv/dvl`	PUB	`stonefish_ros2/msg/DVL`	5 Hz	3-axis bottom-track velocity (m/s) and beam altitude measurements
`/auv/pressure`	PUB	`sensor_msgs/msg/FluidPressure`	10 Hz	Hydrostatic water pressure (Pa) for precise depth estimation
`/auv/display`	PUB	`sensor_msgs/msg/Image`	10 Hz	Forward-looking sonar polar acoustic intensity image (uint8)